

Internship report

Climate Stress Testing Embeddings

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Table of contents

1.	Introduction.....	3
2.	Literature review.....	4
	2.1 History and basics of stress testing and climate stress testing.....	4
	2.2. Main modelling approaches.....	6
	2.3. Application of the machine learning to finance (embeddings).....	8
	2.4 Additional literature.....	9
3.	Materials and methods.....	9
	3.1 Materials (Data)	9
	3.1.1 13-F filings original dataset.....	9
	3.1.2 ESG score and CO ₂ emissions data.....	11
	3.2 Methods.....	12
	3.2.1 Embedding methods.....	12
	3.2.2 Benchmark Methods.....	14
	3.2.3 Stress-testing methods.....	17
4.	Results.....	17
5.	Discussion.....	22
6.	Bibliography.....	24

1. Introduction

This semester, I have been undergoing an internship on the topic of climate stress testing embeddings at Paris Dauphine University – PSL. Having heard about climate stress tests during a Macroeconomics class, I thought that stress tests would be not only an interesting topic, but a chance to align my profile a little bit more with my intended master's degree. During my internship, I read literature on financial economics and sustainability, as well as some literature on Natural Processing Language Models and Artificial Intelligence. I have also downloaded data using API from WRDS and LSEG Workspace and analysed it. In the context of my internship, I have attended as well 2 conferences: 19th Financial Risks International Forum on March 30 and 31 on the topic of hidden risks, and a conference Tech 4 Finance #3: AI and Blockchain on May 11 and 12. One article was central to this internship: a paper by Gabaix et al. "Asset Embeddings" (2025).

The aim of the internship was **two-fold**:

- 1) Make a review of stress-tests and climate stress tests.
- 2) Use the methodological elements of paper by Gabaix, et al. (2025) precisely embeddings to conduct a firm-level stress test.

The novelty lies in the fact that embeddings (a vector representation of an object) have not been very commonly used in finance and economics until now, while in some other areas they became much more prominent (Gabaix et al., 2025). In finance, firm characteristics have been commonly used to explain complex economic outcomes such as relative valuations, the comovement of stock returns, and institutional portfolio decisions, but Gabaix et al. (2025) have advanced the idea that embeddings explain these economic outcomes much better than firm characteristics. Furthermore, they argued that financial institutions, regulators, and central banks could use asset and investor embeddings for risk management and stress testing by generating counterfactual changes in asset and investor embeddings, which would imply counterfactual changes in asset prices in an asset demand system. (Gabaix et al., 2025). The idea was to replicate the finding that embeddings have higher predictive power and then apply them to stress testing.

The main question was: **Do embeddings explain some economic outcomes better than firm characteristics and how embeddings can be used in climate stress testing?**

I have mainly contributed to the preliminary part of the research: literature review and data collection. The plan was to download carbon-related data, download a set of financial characteristics, create embeddings, replicate benchmarks presented in the Gabaix et al. paper

(2025), to check that embeddings explain well certain economic outcomes and then create a vector by taking 2 points in the vector space: that of an investor with brown preferences and that of with green preferences. After creating the vector, the stress-testing should take place. In the current state the benchmarks were replicated, so we could answer the first part of the question, but we haven't responded to the second part yet, as we did not get to the stress testing part.

2. Literature review

2.1 History and basics of stress testing and climate stress testing

The stress tests have already existed in the end of the 20th century, but until the first half of 2007, the interest in stress testing had been circumscribed to risk managers, central bankers and financial supervisors (Quagliariello, 2009). The scale of the financial crisis changed the importance of the stress-testing: mandatory bank stress testing requirements were widely adopted by global regulatory bodies. **Stress test** is a technique that measures the vulnerability of a portfolio, a financial institution, or an entire financial system under different hypothetical events or scenarios (Ong and Jobst, 2020). Any financial system stress test model consists of at 3 or 4 elements (**Figure 1**) (Borio et al., 2014).

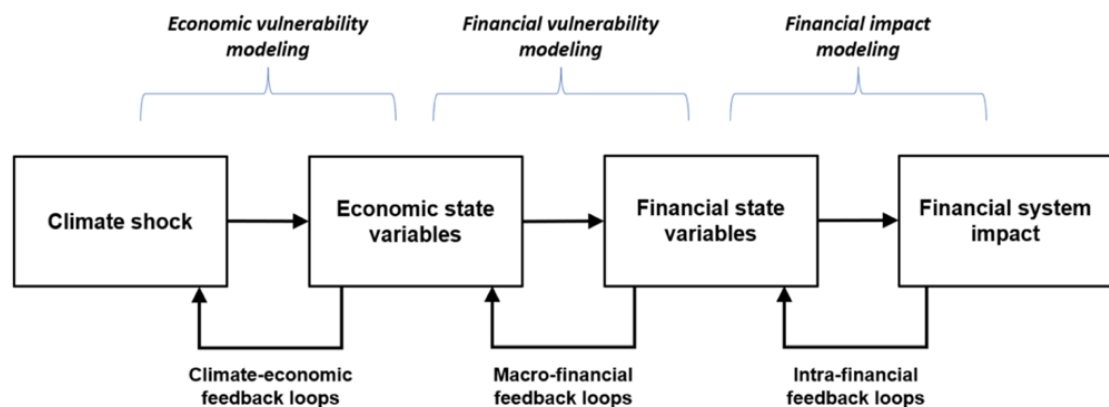
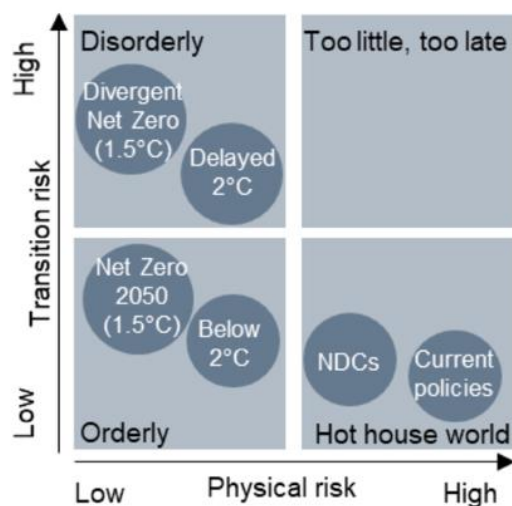


Figure 1. *Conceptual model from climate shock to financial system impact (Reinders et al., 2023)*

1. Shock scenario;
2. **Model** to translate shock variables to macroeconomic variables (also called economic vulnerability).

3. **Model** to translate macroeconomic variables to financial sector variables (also called financial vulnerability).
4. **Model** to apply shocked financial sector variables to financial institutions' balance sheets as well as profit and loss accounts (also called financial impact)

Climate stress tests have emerged from the already existing practice of stress testing. They have appeared not so long ago: the first climate stress test was launched in January 2022 by the European Central Bank (ECB). The reference for the climate stress testing is the Network for Greening Financial System (NGFS) which has produced a set of scenarios. The climate scenario is defined by 2 factors: physical risk and transition risk (Figure 2). **Physical risk** stems from changes in weather and climate (ex. flood results in a decrease of collateral value). **Transition risk** arises from 1) changes in climate policies; 2) emergence of new technologies and increasing affordability of low-carbon technologies; 3) changes in investor and consumer preferences (Deutsche Bank Research, 2023).



There are, thus, 4 main scenarios: **disorderly transition** (low physical risk, high transition risk), **orderly transition** (low physical risk; low transition risk), **hot house world** (high physical risk, low transition risk), and **too, little too late** (high physical risk, high transition risk). There is virtually no use of **too little, too late** scenario in stress testing.

Figure 2. Classification of 4 climate scenarios along two axis (Deutsche Bank Research, 2023)

In climate stress testing, shock scenario and economic vulnerability are one of the most difficult elements to model. While it is possible to use historical data of a previous crisis to model a liquidity shock, no historical data exist that can be applied to climate stress testing. Therefore, complex, multi-level models are used, and models are prone to bias. It is sufficient to remember the Stern vs Nordhaus debate: one parameter can completely change the picture.

There exist several approaches to model the impact of climate change on the economy: 1) macro-financial (traditional) approach; 2) micro-financial approach; 3) non-structural

(direct/empirical) approach; 4) disaster risk approach (Reinders et al., 2023). Models aim to capture financial system in its entirety, but still most practitioners concentrate on some risks. For example, it can be credit risk (potential loss arising from a bank borrower or counterparty failing to meet its obligations in accordance with the agreed terms) or market risk (risk of incurring losses due to adverse movements in financial markets) (CEB, n.d.).

2.2. Main modelling approaches

Macro-financial approach

In macro-financial approach, economic vulnerability can be modelled with a set of different models, where climate shock impacts macro-economic state variables such as GDP, unemployment or inflation, while macroeconomic outcomes can be linked to non-performing loans, credit spreads and bank profitability. For example, a climate stress test of the Colombian banking system conducted by World Bank (2021) uses a set of a set scenarios tailored to Colombia and in line with the recently updated GHG reduction targets. They use computable general equilibrium models (CGE) to model the impact of the transition risk on value added in 83 economic sectors. Then, a credit risk model of the Central Bank of Colombia is used to assess the impact this change in value added has on banks through non-performing loans (Reinders et al., 2021). The non-performing loans can be regressed on a variety of available economic parameters ([Blaschke et al., 2001](#)).

Micro-financial approach

In micro-financial approach, climate shock affects micro-economic state variables at firm or asset level, such as earnings, enterprise value, and leverage. To go from model to go from **climate shock** to **microeconomic state variables**, discounted cash flow analysis is one of the most recently used models. In this approach is based on carbon prices are used as negative cash flows, based on Merton's ideas on equity. One clear advantage is that this model is applicable to asset level, but it is only limited to modelling the effects of carbon taxation. Then, regressions are used to assess evolution in financial parameters.

Non-structural approach

Among the approaches presented in this literature review, the non-structural approach will be the closest approach to the one we have adopted: in this approach the climate shock affects

directly the financial parameters, and everything that happens in the economy is considered to be a black box (**Figure 3**).

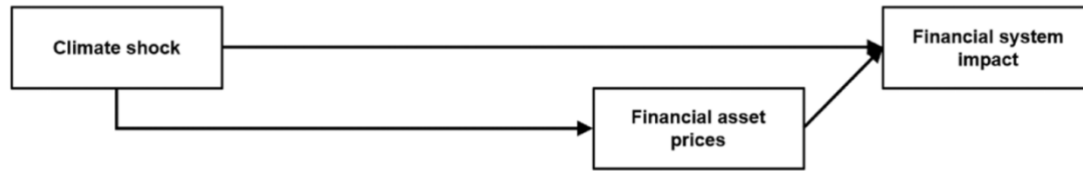


Figure 3. *Schematic representation of the non-structural approach (Reinders et al., 2023)*

We can provide 2 papers exemplifying the non-structural approach: Battiston et al. (2025) and Koijen et al. (2024). The first one conducts a stress test on large Euro Area Banks, while the second one conducts stress test on US investors.

In the model proposed by Battiston et al. (2025) the fact that an event can induce not one but several rounds of losses in the financial system is considered. Two round of losses can be induced by climate change in the model: first round accounts for the losses in banks' equity (value of an asset after deducting its liabilities) due to direct exposures to shocks (it is supposed that all equity from industries most exposed to the transition risk is lost), while the second round includes indirect losses in banks' equity due to the devaluation of counterparties' debt obligations on the interbank credit market induced by the loss of equity (Battiston et al., 2017). The risk can propagate through direct linkages between banks; in this case it is possible to divide assets by those hold by institutional and non-institutional actors to better capture the contagion effect. The exposures of each financial actor can be decomposed along the main types of financial instruments: equity holdings (for example, ownership shares including both those tradable on the stock market and those non-tradable), bond holdings (for example, tradable debt securities) and loans (for example, non-tradable debt securities).

The paper by Koijen et al. (2024) investigates how climate-induced shifts in asset demand due to sustainable investing or changes in climate regulation (stakeholder and regulatory risk) affect equity prices on the level of individual investors, namely the US institutional investors (investment advisors, hedge funds, pension funds and insurance companies, private banking, and brokers). The authors use firm characteristics and model the trend toward sustainable investing as a shift in asset demand toward greener stocks by an increase in the elasticity of

demand to the environmental score. This shift in its turn induces changes that result in higher valuation of green stocks and lower valuation of brown stocks.

Disaster risk approach

Disaster risk models are used in the insurance industry and are applied to financial sector outcomes, through variables such as economic damage and total factor productivity (TFP). Relationship between damages and financial variables is defined based on existing historical data. These models have hazard module (potential adverse events, i.e. flood), a vulnerability module (damage function), an exposure module (location, characteristics and value of objects), and a financial loss module (i.e. insured losses).

2.3. Application of the machine learning to finance (embeddings)

As for the embeddings, there exists little literature on their application to finance except Gabaix et al. as it was previously stated, and no literature on their application to stress testing. An embedding is a vector representation of a word that captures its meaning, extracted from text data by natural language processing (NLP). Embeddings capture similarities between words, sentences, and documents and can do math with words. In a classic example, the embeddings for “king”, “man”, and “woman” are used to compute “king – man + woman”, which produces an embedding closest to “queen” (Gabaix et al., 2025). In the financial context, an embedding is a vector representation of an object in the financial fillings’ documents (Gabaix et al. 2025). Portfolio holdings contain all the relevant and necessary information to generate embeddings. The asset embeddings capture differences in expected profitability or risk exposure across assets. In other words, if two assets have similar embeddings, they have similar profitability and are prone to same risks. The investor embeddings capture heterogeneity in preferences for the asset embeddings across investors. Investor embeddings represent particular interest for us, as one of the steps of the research is the construction of a green-brown vector using investor embeddings.

In our work, we had to replicate benchmarks from the paper of Gabaix et al. (2025) before proceeding to stress testing, and the replication crisis has affected finance literature the same as many of other fields. There exist two hypotheses from recent literature on replication crisis in finance: there is no internal validity due to coding errors or faulty statistics in the research process (H1), and there is no external validity due to p-hacking (H2). Nevertheless, these 2 hypotheses were recently disproved (Jensen, Kelly and Pedersen, 2023).

2.4. Additional literature

We have mentioned that the risk can propagate through direct linkages between financial actors, but it can also propagate through fire sales (Greenwood et al., 2014). When a bank experiences a negative shock to its equity, a natural way to return to target leverage (central banks are required to possess liquidity representing a certain percent of their liabilities) is to sell assets. If potential buyers are limited, then asset sales depress prices, because banks might prefer to sell at a lower price than the worth of the asset rather than not to sell anything. In this case one banks' sales impact other banks with common exposures, and this contagion effect adds up across the banking sector (Greenwood et al., 2014).

There exists as well reverse stress testing that addresses the inverse of conventional stress testing by asking which configurations of risk drivers could plausibly generate a pre-specified adverse outcome. The focus of such reverse stress tests is the joint severity and plausibility of scenarios leading to failure states (Hurlin et al., 2026).

During the 19th Financial Risks International Forum, some papers have been presented that treat climate as well and sustainability in finance, like a paper by “Quantifying Climate Risk Premia” (Melin and Zhang 2025), but as they were tangential to the research, I will omit their summary.

3. Materials and methods

3.1. Materials (Data)

3.1.1 13-F filings original dataset

The initial dataset contained data for 29 unique quarters, 7147 unique funds/investors, and 5052 unique stocks/firms, which results in a dataset of 30 million rows. It was retrieved from WRDS and the data originated from 13-F fillings. It was possible to get from this dataset information on the number of shares and holding value of various asset for different US funds. The earliest date for which data was available in 2017Q4 and the latest is 2024Q4. All of the funds we had were mutual funds, contrary to the FactSet dataset from the original paper including a broader range of institutional investors. Mutual fund is an open-end investment company (a company that pools investors' money and spreads it across a wide range of investments, such as equities or fixed-interest securities). The combined holdings the mutual fund owns are known as its portfolio, which is managed by an investment adviser. Investors in mutual funds buy their shares from, and sell/redeem their shares to, the mutual funds themselves or through investment professionals like brokers or investment advisers.

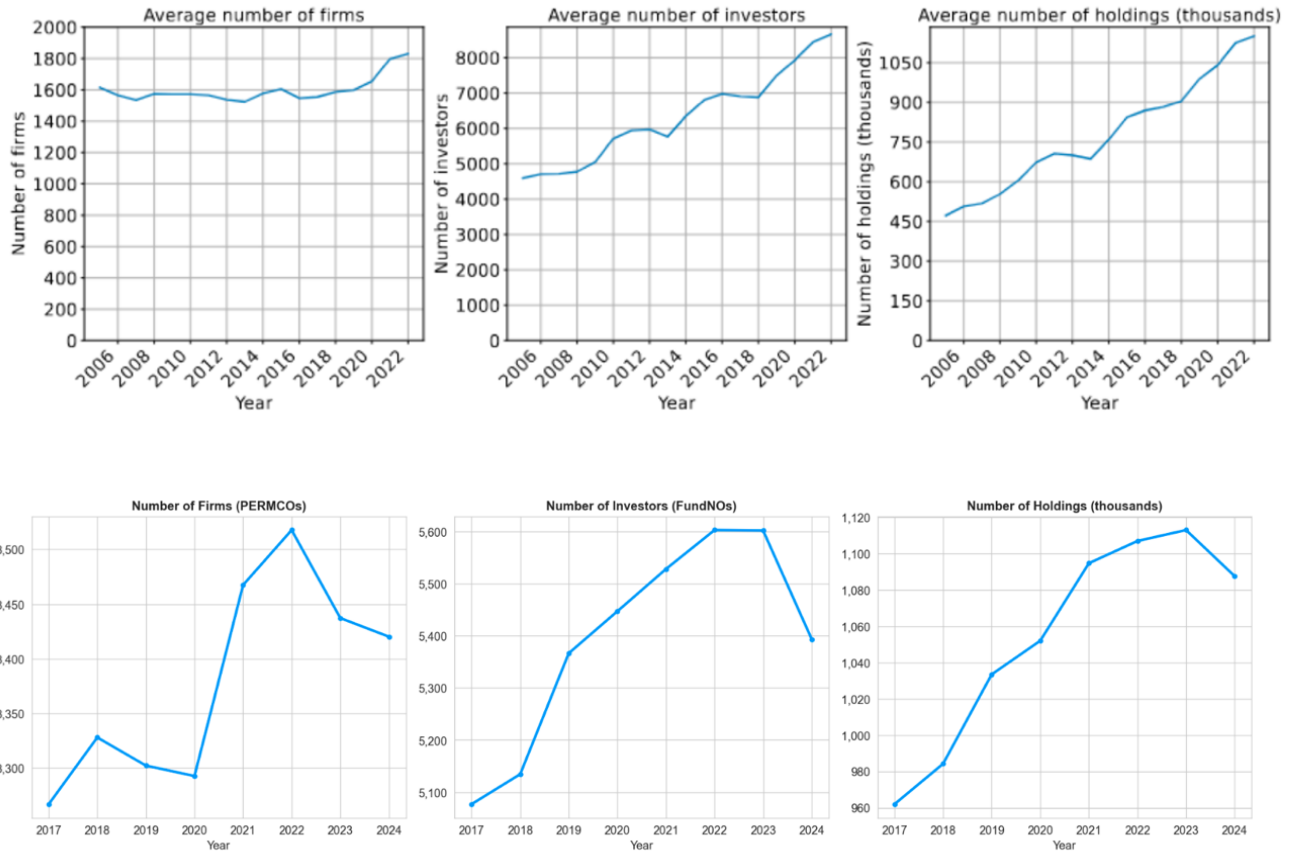


Figure 4. Comparison between our data (upper panel) and data from Gabaix et al. (lower panel). We had shorter time span (2017-2024) compared to that of the paper (2006-2022), but we had data for 2023 and 2024 contrary to the paper. For 2020-2022, a) we had a larger coverage of firms: below 3300 – above 3500 compared to less than 2000, b) smaller coverage of investors (which was expected given that we only had mutual funds, while FactSet dataset had more investor types). The average number of holdings is quite close for both datasets.

As the dataset was very large, I did some grouping by the average holding value throughout the period and chose top 20 fund families to plot (**Figure 5**). Even though there are numerous unique funds, many of them can be part of the same fund family. For instance, there are more than 75 unique firms in the Vanguard Group, which is one of the largest family funds (**Figure 6**). At the time I did not have access to the LSEG API, so I tried the manual grouping by first name. However, it did not prove very successful, due to the fact the funds from the same family don't always share the same starting word. For instance, fund names starting with "iShares" and "BlackRock" belong to the same family BlackRock. Therefore, as I was out of

options, until I succeed to use yahoo finance API (it was the only available open-access API at the moment).

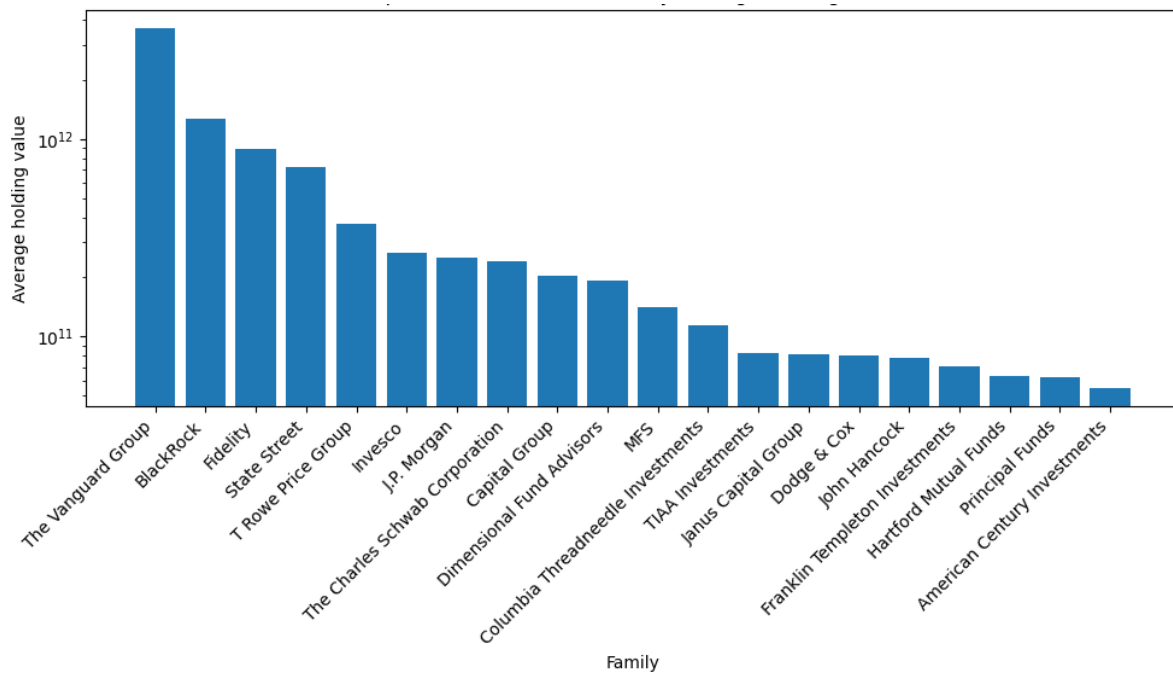


Figure 5. Top 20 Families of investors by average holding value throughout the period (2017Q4-2024Q4). It is visible that all top 20 families by average holding have their values in a range of slightly below 10^{11} and slightly above 10^{12} (thousand dollars). So, there is not much variability in holdings among the largest families.

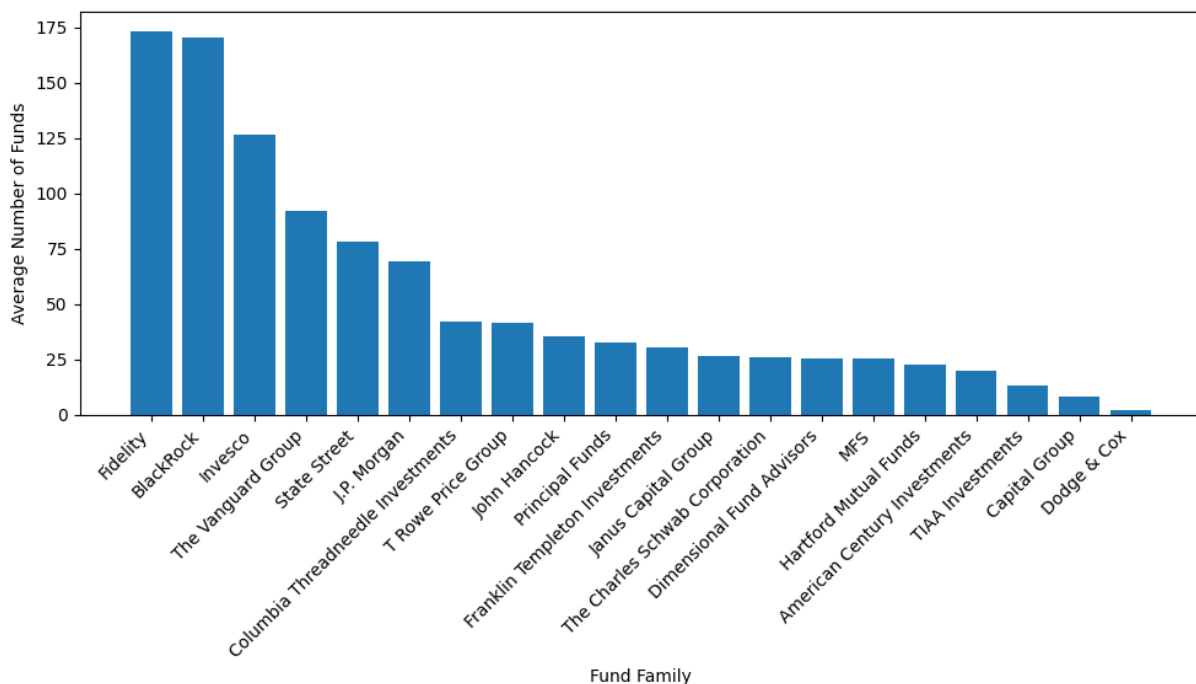


Figure 6. *Average number of funds per fund family throughout the period (2017Q4-2024Q4)*

Nevertheless, this dataset alone was insufficient for conducting next steps: benchmarking replication and stress testing. For stress-testing, carbon-related data was necessary.

3.1.2. ESG score and CO₂ emissions data

Merging the initial dataset and the Institute Louis Bachelier (ILB) Corporate GHG Emission Estimations (COGEM) dataset, I had to conclude that the COGEM database had intermediate coverage in terms of Total CO₂ emissions (50-60% correspondence between data on stocks emissions and stocks hold by top 20 family funds). However, I later discovered that it was not the problem of the database. Even in LSEG Workspace, where it is possible to access data of practically any company on the stock market, data on CO₂ emissions is much less present than data on the ESG score. The ESG score was available in 2023 for around 76% of firms, while the CO₂ emissions were available for less than 50% of them.

From the LSEG workspace, various data was downloaded on ESG and carbon emission: ESG score, ESG controversies score, separate E, S and G score, ESG combined score, Total CO₂ Emissions, Scope 1 Emissions, Scope 2 Emissions, Scope 3 Emissions and Total CO₂ emissions to revenue. There was also information on the ESG auditor's name.

3.2. Methods

3.2.1 Embedding Methods

Embeddings can be obtained from portfolio holdings through various models: recommender models, shallow network models and transformer models. All of them are Natural Language Processing Models (NLP). Natural language processing is a subfield of computer science and artificial intelligence (AI) that uses machine learning to enable computers to understand and communicate with human language. We made each model generate embeddings with 4, 32, 64 and 128 dimensions.

We first need to determine the relation between portfolio holdings and embeddings. We start with traditional demand equation, where demand depends on price and quantity.

$$Q_i = Q_i(P, x)$$

However, in the paper Q is an investor's vector of asset demand, P - the vector of prices and x - the matrix of other asset characteristics (embeddings). This formula describes demand for

assets as dependent not only on its own price and characteristics but as well on the price of other assets and characteristics of other assets (General Equilibrium Model). Then, there is a market clearing condition,

$$\sum_i Q_i = S,$$

where S denotes the vector of asset supplies. Supply depends as well on price and characteristics of an asset, $S(P, x)$.

So, if there is only one solution $P = b(x)$, then reduced-form demand depends exclusively on asset characteristics.

Investor's dollar holding of an asset is modelled as:

$$h_{ia} = \kappa_i + (1 - \zeta_i)p_a + v_{ia}$$

κ_i - a constant that scales with the investor's wealth and ζ_i - the price elasticity of demand for individual assets, and v_{ia} - a demand shifter, p_a - price of the asset.

The final equation we come to is the equation of reduced-form demand (an equation that illustrates how an endogenous variable responds to changes in predetermined or exogenous variables, not considering interdependencies between the variables):

$$h_{ia} = \lambda_i' x_a + \delta_i + \delta_a + \epsilon_{ia} \quad (1)$$

Where h_{ia} is log dollar holding $\lambda_i' x_a$ is the product of investor and asset embeddings, and δ_i, δ_a are fixed effects. Therefore, we can represent investor's log dollar holding of an asset as the dot product of the investor embedding and the asset embedding.

Recommender models

In a simple recommender system, we applied Singular Value Decomposition to a demeaned holdings matrix (holding matrix received with the help of *equation 1* where the mean is subtracted from each individual holding value) to estimate asset and investor embeddings. There are several variants of the recommender system that were proposed in an original paper. As we already could see that 2 best performing are RS-L-Min and RS-L-0, so we used only these 2. In RS-L-Min zero positions (absent data) are replaced with minimal dollar holding for the given investor, while in RS-L-0 the zero positions are replaced by zeros.

Shallow network models

Word2Vec, which is a shallow network model, exploits the fact that investors assign similar portfolio weights to assets with similar asset embeddings, according to the portfolio theory underlying equation. To train this model, for each investor, assets are ranked in decreasing order of portfolio. Word2Vec then assigns a probability distribution over the masked asset, conditional on the average asset embedding in the context window (10 assets). In this way, the model generates asset embeddings. We can as well generate investor embeddings ordering by the ownership share.

Transformer models

Lastly, transformer models are represented by OS-BERT and PS-BERT model. The advantage of these models is so called attention mechanism (Prince, 2023). These models evaluate context in which they found a word. For instance, while other models will interpret “Apple” in “Apple has developed new Mac” and “apple” “I bought an apple on the market” as the same item and will assign the same embedding, transformer models will distinguish them. To use the transformer models, it was necessary to convert numerical values of portfolio holdings to discrete words that these models use as inputs. To do so, we ordered the assets in an investor’s portfolio in a decreasing order of portfolio weights and simply represent the name of each asset as a token (OS-BERT), and we ranked investors by the ownership share (PS-BERT). We start by assign to a certain asset an embedding, and throughout the training by comparing asset’s ranking, the model assigns embeddings to other assets. The same goes for investors.

I didn’t work on writing the code to generate embeddings, as I didn’t have the necessary knowledge, and the code was written by PhD student Renzhi Liu and shared with me. So, I could examine the code, and I could manipulate snippets of the code to see how it was working.

3.1.3. Benchmark methods

To replicate benchmarks to see if embeddings explain well economic outcomes (relative valuations, comovement of stock returns, institutional portfolio decisions) and outperform firm characteristics, I had to download some additional data in the WRDS with the use of an API: firm characteristics: market beta (measures the volatility of a security or portfolio compared to the broader market), asset growth, profitability, dividend/assets, log ME (market equity), log B/M (market-to-book equity), momentum (the rate at which a security's price changes,

allowing traders to profit from sustained trends), and monthly returns. The 3 benchmarks below are chosen because they satisfy three important criteria. First, the portfolio holdings data are informative for these benchmarks. Second, these benchmarks are at the heart of asset pricing and portfolio theory, Third, and most distinctively, these benchmarks require only a single quarter of data (Gabaix et al., 2025).

Relative Valuation Benchmark

It is possible to access the valuation of an existing firm or predict the valuation of a new firm through a comparison with similar firms matched on characteristics. Instead of firm characteristics Gabaix et al. and we used asset embeddings. Market-to-book equity was taken as a proxy for valuation. Market equity is a product of price per share and outstanding shares, while book equity is the true value of the company. The data we had was divided in training (80%) and testing (20%) data.

Step 1. Regress for each quarter log market equity (p_{at}) on log book equity (b_{at}); the residual ($p_{at}^\perp$) is the valuation measure:

$$p_{at} = \gamma_t b_{at} + \alpha_t + p_{at}^\perp$$

Step 2. on 80% training firms, perform ridge regression of valuation (a statistical method used to reduce errors caused by multicollinearity for linear regressions model (IBM; n.d.)) on embeddings:

$$p_{at}^\perp = \beta'_t x_{at} + \delta_t + \epsilon_{at}$$

x_{at} – asset embeddings, δ_t – fixed effect, ϵ_{at} – error term

Step 3: Calculate out-of-sample (OOS) R^2 on 20% test firms, averaged over quarters to compare a model (embeddings) with a baseline prediction model (characteristics):

$$R^2 = 1 - \frac{1}{T} \sum_{t=1}^T \frac{Var(p_{at}^\perp - \hat{\beta}'_t x_{at} - \hat{\delta}_t)}{Var(p_{at}^\perp)}$$

Where $Var(p_{at}^\perp - \hat{\beta}'_t x_{at} - \hat{\delta}_t)$ – explained variation of the model, $Var(p_{at}^\perp)$ – total variation of the model, T – quarters. To calculate the baseline R^2 four firm characteristics were used: market beta, asset growth, profitability, dividend/assets.

Stock Comovement Benchmark

A large literature in asset pricing finds variation in average stock returns and comovement of stock returns along firm characteristics. So, we tried to replicate the use of embedding to explain comovement of stock returns. The underlying assumption is that if assets are similar, their returns would move in a similar way, due to the exposure to the same factors. The asset embeddings in quarter $t-1$ to explain the monthly stock returns in month m of quarter t .

Step 1. For each month m in quarter t , regress monthly returns on lagged embeddings (quarter $t-1$), on 80% training firms:

$$r_{am} = \beta'_m x_{a,t-1} + \delta_m + \epsilon_{am}$$

β_m - vector of monthly factor realizations; δ_m – fixed effect

Step 2. Calculate out-of-sample (OOS) R^2 on 20% test firms, averaged over months:

$$R^2 = 1 - \frac{1}{M} \sum_{m=1}^M \frac{Var(r_{am} - \hat{\beta}'_m x_{a,t-1} - \hat{\delta}_m)}{Var(r_{am})}$$

To calculate the baseline R^2 six firm characteristics are used: market beta, log ME, log B/M, asset growth, profitability, momentum.

Managed Portfolio Benchmark

This benchmark tests the core prediction that investors assign similar portfolio weights to assets with similar asset embeddings.

Step 1. Predict a masked asset. As Word2Vec and PS-BERT were already trained to predict masked assets, they assign prediction probabilities directly from their training objectives.

As for the recommender system, this formula is used:

$$Pr(b_i(k) = a | A_i(k)) = \frac{\exp(\lambda'_i x_a + \delta_a)}{\sum_{b \notin A_i(k)} \exp(\lambda'_i x_b + \delta_b)}$$

$b_i(k)$ - the k^{th} largest masked position; $A_i(k)$ – set of all other assets in the investor's portfolio, δ_a, δ_b – fixed effects.

Step 2. McFadden pseudo- R^2 is calculated using this formula

$$R^2 = 1 - \frac{\sum_{b_i(k) \in B} \ln \Pr(b_i(k) = a | A_i(k))}{\sum_{b_i(k) \in B} \ln(1/|A_i(k)|)}$$

The performance measure is the average log likelihood of correct predictions across investors, relative to the log likelihood of random guesses.

3.1.4. Stress-testing methods

Even though we didn't get to the stage of generating changes in the embeddings, I have read the relevant literature from Koijen and Yogo, (2019). Koijen and Yogo proposed an efficient algorithm for computing the equilibrium price vector in any counterfactual experiment: the Newton's method:

$$p_{m+1} = p_m + \left(I - \frac{\partial f(p_m)}{\partial p'} \right)^{-1} (f(p_m) - p_m)$$

Where p_{m+1} - new equilibrium price, p_m – estimation of current equilibrium price; I - identity matrix, $\frac{\partial f(p_m)}{\partial p'}$ - Jacobienne, $(f(p_m) - p_m)$ – error term

Normally, this equation would be useful in future steps of the research.

4. Results

Relative Valuation Benchmark

For the relative valuation benchmark, we can conclude that our results are comparable to that of Gabaix et al. Firstly, because RS-L-Min is the best model for relative valuation both in our tests (with $R^2 = 0.722$ for $K = 128$) and in the paper. Furthermore, all models that generate embeddings perform better than firm characteristics for 32 dimensions and more. We can see that the performance scales with dimension K for all models (RS-L-min, RS-L0, W2V and OS-Bert). Nevertheless, we had some differences, especially with OS-BERT predicting little relative valuation at 4 dimensions. In our tests $R^2 < 0.10$ in Gabaix et al. paper $R^2 \approx 0.3$.

For firm characteristics, we took $R^2 = 0.125$ as mentioned in the paper. Therefore, embeddings explain 5-6 times more variation than firm characteristics.

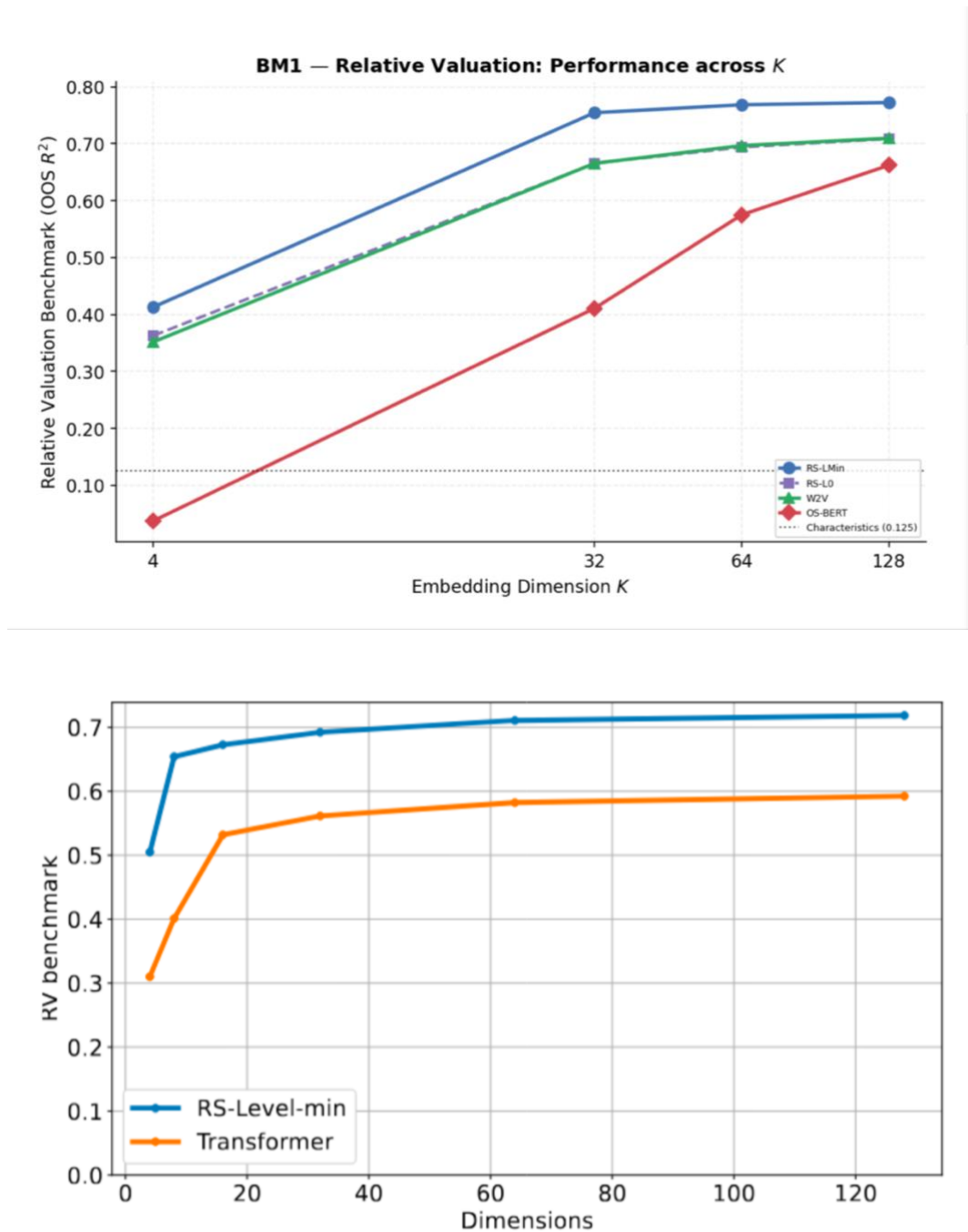
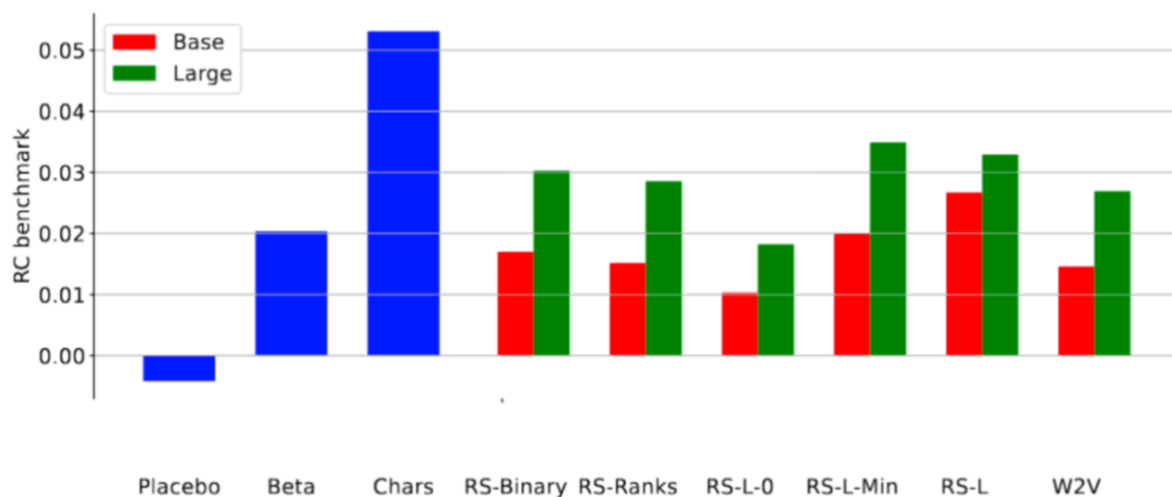
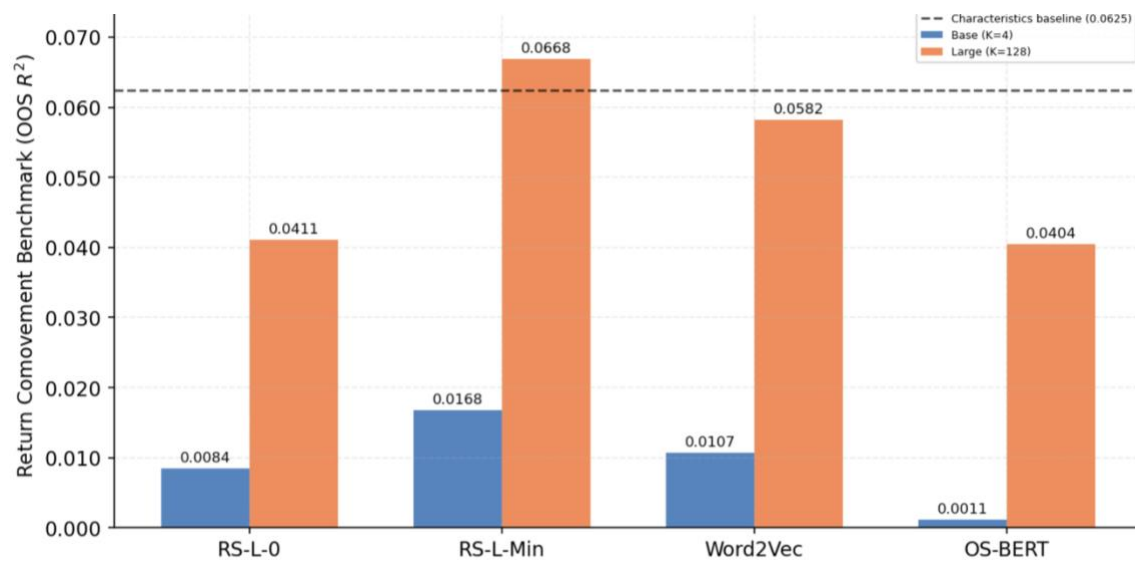


Figure 8. R^2 values in our tests (upper graph) and in the paper (lower graph). In the paper R^2 for relative valuation benchmark is called RV.

Stock Comovement Benchmark

For the stock comovement benchmark, we observe similar situation to that described in the paper: firm characteristics explain stocks comovement better than embeddings. Embeddings start to explain the stocks comovement better only at high dimensions, and it is only one model RS-L-Min at $K = 128$ that succeed to outperform the firm characteristics.



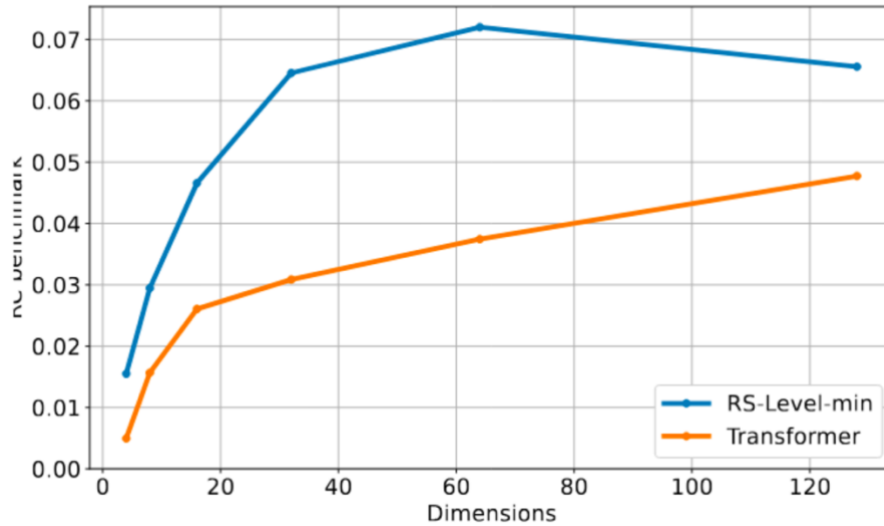
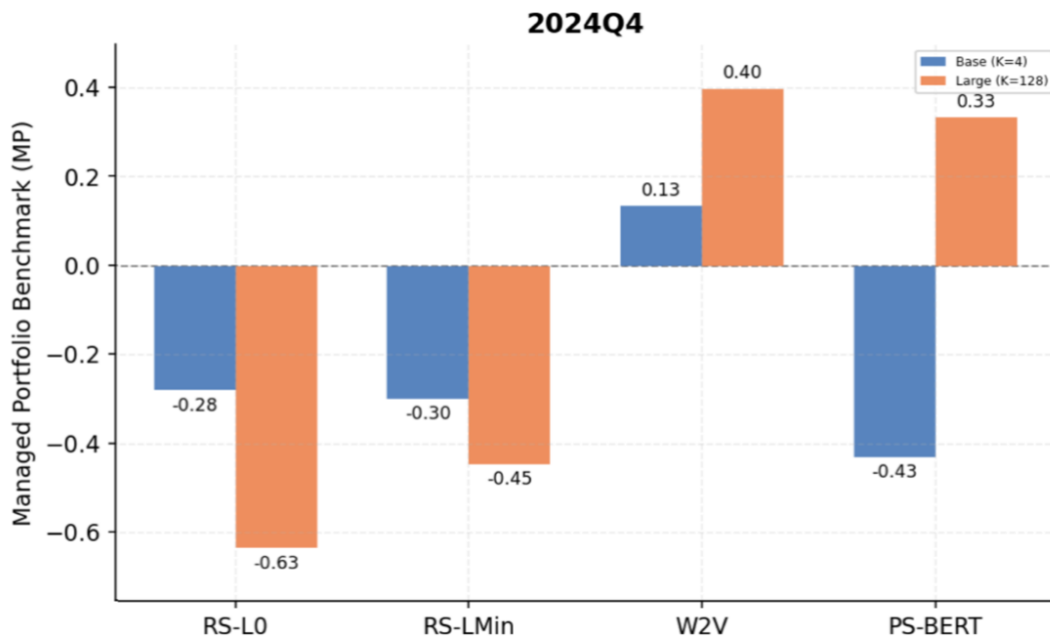


Figure 9. R^2 values in our tests (upper graph) and in the paper (middle and lower graph). In the paper R^2 for stocks comovement benchmark is called RC.

Managed Portfolio Benchmark

We took 2024Q4 quarter to conduct the tests. The best model is W2V K=128, $R^2 = 0.398$; while PS-BERT comes second; RS models perform poorly on this benchmark (negative MP). Masked-prediction models (W2V, BERT) dominate as expected, as they are trained on exactly this task (prediction of a masked asset). We didn't calculate the R^2 for the characteristics due to the lack of time.



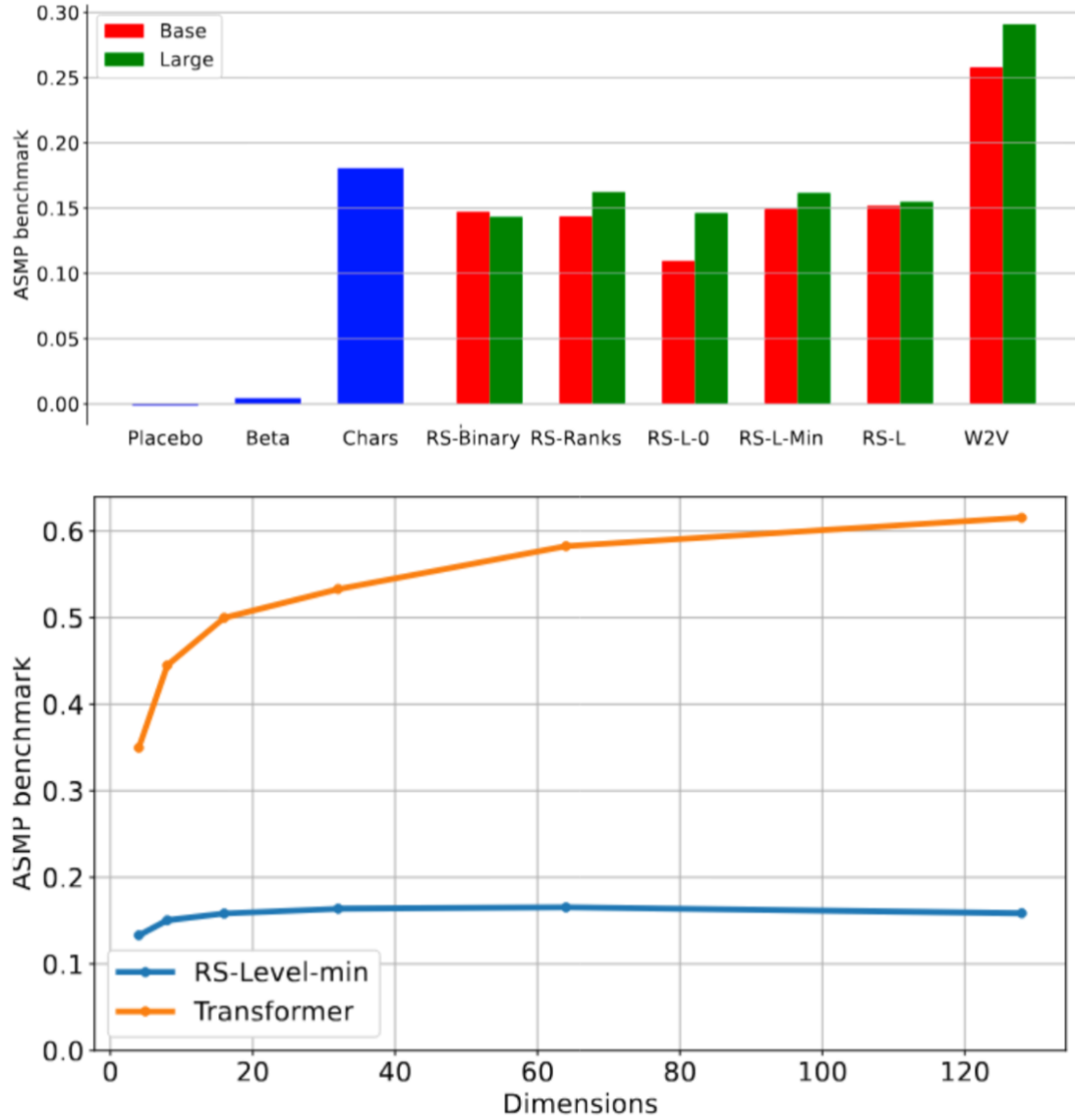


Figure 9. R^2 values in our tests (upper graph) and in the paper (middle graph compares recommender model, characteristics and W2V and lower graph compares recommender system RS-L-Min and transformer model PS-BERT). In the paper R^2 for managed portfolio benchmark is called MP.

	BM1 (RV)	BM2 (RC)	BM3 (MP)
Best model (ours)	RS-L-Min K128	RS-L-Min K128	W2V K128
Best value (ours)	0.772	0.0668	0.398
Characteristics	0.125	0.0625	—
Paper ranking	RS-L-Min > OS-BERT	RS-L-Min > OS-BERT	PS-BERT > W2V
Our ranking	same	same	W2V > PS-BERT

Table 1. *Summary of the results*

Therefore, embeddings explain the economic variables such as relative valuation, stock return comovement and portfolio investment decisions and outperform firm characteristics on some outcomes. We succeeded to replicate the ranking of models of Gabaix et al. for 2 benchmarks, but for one benchmark we have established a different ranking.

5. Discussion

As for the results, the difference in the RV in the Relative Valuation Benchmark between OS-Bert in the tests run by Gabaix et al. and our tests, some differences in RV value of firm characteristics, slightly above 0.5 for Gabaix et al. and 0.668 for us in stock comovement benchmark, and difference in ranking of models for managed portfolio, can be due to the fact that our data was smaller: we had only mutual funds, while as it was previously mentioned FactSet dataset of Gabaix et al. contained as well amongst others pension funds and hedge funds. As the replication of benchmarks was quite successful, it is possible to move to the next step: stress testing itself.

Another important finding that is not present in the results but is more of an observation comes from my work with downloading ESG indicators and CO2 emissions. There seemed to be a clear preference for reporting ESG score rather than CO2 total emissions or scope emissions. My explanation is that ESG score is more opaque while CO2 total emissions are a clearer measure of the environmental performance. In Environmental Pillar Emissions, Innovation and Resource Use are covered. It is worth noting that Emission use data of CO2 emissions, although the number of companies who have ESG score and those who have CO2 emissions reported are quite disparate. In Social pillar a range of categories is included: community, human rights, product responsibility and workforce. As for the governance, available is data on CSR strategy, management and shareholders. It is interesting to look at weights assigned to each pillar: they differ for different companies. I haven't found the explanation in the methodology available on the website. For instance, for Apple Inc Environmental Pillar Score has a weight of 23.0%; Social Pillar Score the weight of 49.3% and Governance Pillar Score has the weight of 27.6%. For Nvidia Environmental Pillar Score has a weight of 31.2%; Social Pillar Score the weight of 42.9% and Governance Pillar Score has the weight of 26.0%. The score values can vary from A+ to D-. The methodology refers to the

weight allocation in this manner “To calculate the overall pillar and ESG scores, category weights per industry are applied using data-driven and objective logic”.

Benefits

During the internship, I have acquired theoretical and practical knowledge. Not only I got acquainted with the works of NGFS, but I rediscovered some things learnt in lessons. For instance, the use of the Nordhaus model and other models of the same type to relate physical and transition risks to the economy. I learnt to use APIs for LSEG and WRDS, which is quite useful, because the process is rather complicated: not only documentation is quite large, but in WRDS SQL is required instead of Python. I particularly appreciated work with the ESG statements on LSEG workspace: it was insightful to see how ESG data is reported, who are the main ESG auditors, and that weights assigned to E, S and G differ. The fact that some companies (Nvidia) can have an A in ESG score and D- in ESG controversies score reminded me once again about the pitfalls of indicators and of the Measurement theory classes I had in my first year of the bachelor's degree. I also got more acquainted with financial variables and even went studying a page on portfolio theory (it was not that important to describe in detail in the report) but quite insightful. I also read a huge corpus of articles on financial economics, stress testing and expanded my knowledge in NLPs.

Difficulties

What concerns the difficulties I have faced in my work, most of them were related to the data, as I mainly worked with it. The dataset was very big, so I had to spend a lot of time doing coding operations, and it resulted in a crashed kernel 2 or 3 times. The large amount of present data complexified the description of the dataset and made manual work impossible: it was necessary to automate many tasks. Another issue was the quality of data and correspondence between WRDS and LSEG databases: the query of the LSEG database using API was complexified by the fact that in WRDS dataset there were many abbreviations and a fair number of mistakes. To illustrate, I have spent a great deal of time trying to understand why there was no ticker for “LUMEN TECHNOLOGIES INC” in the LSEG Workspace, to remark later that later “N” was absent from the word “TECHNOLOGIES”. There have been at least a hundred cases like this. And there was no way to correct this except by hand.

Moreover, it is sometimes complex to write the most efficient code for the APIs: at first, I queried them one by one, then, I discovered that I could do so in batches, and finally, I found

out that it is possible to use several so-called workers, which permit to simultaneously query several batches. This last discovery has greatly facilitated the work but was made by me a bit late.

Unfortunately, I couldn't work as much with stress testing as I would like to due to time constraint and to the fact that some steps were too complex for me to perform on my own, but I gained a lot of useful theoretical and practical knowledge.

Applications

I hope that the data I downloaded will be useful for future research on this topic which would be continued. One of the main issues of the model according to the authors themselves is the fact that asset and investor embeddings are latent vectors that do not have a direct economic interpretation. Gabaix et al. proposed feeding the fillings in an LLM for interpretation, but it is interesting to see later if any other methods could work.

Remerciement

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